

Classic Riddler

25 March 2022

Riddle:

The astronomers of Planet Xiddler are back at it. They have developed a new technology for measuring the radius of a planet by analyzing its cross sections.

And so, they launch a satellite to study a newly discovered, spherical planet. The satellite sends back data about three parallel, *equally spaced* circular cross sections which have radii A , B and C megameters, with $0 < A < B < C$. Based on these values, the scientists calculate the radius of the planet is R megameters. To their astonishment, they find that A , B , C and R are all whole numbers!

What is the smallest possible radius of the newly discovered planet?

Solution:

I calculate that the smallest possible radius is 65 megameters. This occurs in the Pythagorean triples (60,25,65); (56,33,65); and (52,39,65). The side lengths of 60, 56, and 52 are equally spaced, so the cross sections have radii 25, 33, and 39. Interestingly, I also found the Pythagorean triple (63,16,65), although it doesn't fit in with the pattern. So my solution is **65 megameters**.